HARVESTLINK PLATFORM

Agile Functional Specifications & User Story Matrix

- **Document Title:** Software Requirements Specification (Agile Backlog Specification)
- **Document Version:** 1.0
- **Author:** Dilshani Rathnayake (Lead Business Analyst & Agile Project Coordinator)
- **Project Phase:** Sprint Grooming & Detailed Scoping
- **Target Domain:** Agribusiness Supply Chain & InsurTech
- **Date:** October 2026

Version	Date	Author	Description of Changes	Approved By
v1.0	October 2026	Dilshani Rathnayake	Baseline 6 Core User Stories with Gherkin Acceptance Criteria	Technical Lead / Scrum Master

Persona 1: Sunil Perera (Smallholder Producer)

- **Role:** Independent Farmer (Vegetables & Cash Crops, Badulla/Nuwara Eliya belt)
- **Pain Point:** Unpredictable broker pricing at open economic centers and crop destruction from sudden heavy rains.
- **Primary Goal:** Secure pre-harvest forward contracts with price guarantees and access fast parametric insurance.

Persona 2: Chamara Silva (Commercial Procurement Manager)

- **Role:** Enterprise Agri-Buyer (Retail Supermarket Chain)
- **Pain Point:** Quality discrepancies and lack of field visibility before harvest delivery.
- **Primary Goal:** Contract future crop batches directly from farmers and hold secure funds in escrow.

Persona 3: Saman Wickramasinghe (Assessor / System Admin)

- **Role:** Agricultural Claims Assessor & Platform Moderator
- **Pain Point:** Fraudulent weather claims and lengthy physical verification delays.
- **Primary Goal:** Audit weather indices, approve automated payouts, and resolve trade disputes.

2. Core User Stories & Gherkin Acceptance Criteria

Story ID: US-01 — Land Parcel & Crop Cultivation Registration

- **User Story:**
  As a Registered Smallholder Farmer,
  I want to register my farm plot dimensions, crop variety, and expected planting date,
  So that my production capacity becomes visible to prospective institutional buyers.
- **Acceptance Criteria (Gherkin Syntax):**
  - **Scenario 1: Successful plot creation**
    - **Given** the farmer is authenticated on the HarvestLink mobile app and opens the "Add Land Plot" screen
    - **When** the farmer enters valid land extent (in acres), selects a crop variety, sets an estimated planting date, and confirms GPS location
    - **Then** the platform saves the parcel record, assigns status "Verified Plot", and displays a success notification.
  - **Scenario 2: Validation failure on missing required fields**
    - **Given** the farmer is filling out the land registration form
    - **When** the farmer submits the form without providing the crop variety or land size
    - **Then** the system highlights the missing input fields in red, disables the submission button, and displays an inline error message.

Story ID: US-02 — Forward Supply Contract Offer Creation

- **User Story:**
  As a Registered Farmer,
  I want to publish a pre-harvest forward supply contract specifying a floor price per kilogram,
  So that I lock in guaranteed revenue prior to cultivation and eliminate intermediary broker exploitation.
- **Acceptance Criteria (Gherkin Syntax):**
  - **Scenario 1: Publishing a valid forward offer**
    - **Given** the farmer has an active "Verified Plot"
    - **When** the farmer specifies target harvest yield (kg), minimum floor price per kg (LKR), and estimated delivery week, and clicks "Publish Offer"
    - **Then** the contract status changes to "Open for Reservation" and is indexed on the Buyer Marketplace board.
  - **Scenario 2: Invalid floor price entry**
    - **Given** the farmer is configuring a forward contract offer
    - **When** the farmer inputs a floor price equal to or below zero LKR
    - **Then** the system rejects the input and prompts "Floor price must be greater than zero".

Story ID: US-03 — Buyer Contract Reservation & Escrow Deposit Lock

- **User Story:**
  As an Enterprise Procurement Buyer,
  I want to browse active forward supply listings and reserve a batch by committing an upfront escrow deposit,
  So that I secure future vegetable volume for my retail supply chain.
- **Acceptance Criteria (Gherkin Syntax):**
  - **Scenario 1: Successful escrow commitment**
    - **Given** a forward contract is listed with status "Open for Reservation"
    - **When** the buyer selects "Reserve Batch" and authorizes an upfront 20% escrow deposit from their corporate balance
    - **Then** the platform locks the escrow funds, updates the contract state to "Reserved / Active", and sends an SMS/push notification to the farmer.
  - **Scenario 2: Insufficient escrow wallet balance**
    - **Given** the buyer attempts to reserve a contract batch
    - **When** the required 20% deposit exceeds the buyer's available account balance
    - **Then** the transaction is blocked, the contract remains "Open", and the buyer is prompted to top up their account funds.

Story ID: US-04 — Cultivation Milestone Tracking via Photo Verification

- **User Story:**
  As a Smallholder Farmer with an active contract,
  I want to capture and upload field photos at critical growth intervals (Planting, Flowering, Pre-Harvest),
  So that the buyer and insurer have verifiable proof of cultivation progress and crop health.
- **Acceptance Criteria (Gherkin Syntax):**
  - **Scenario 1: Successful milestone update submission**
    - **Given** a contract is in "Reserved / Active" status and a scheduled growth milestone is due
    - **When** the farmer captures a field photo via the mobile app and submits it with optional progress notes
    - **Then** the platform records the upload timestamp, marks the milestone as "Completed", and alerts the assigned buyer dashboard.
  - **Scenario 2: Low-quality or unreadable upload attempt**
    - **Given** the farmer initiates a milestone photo submission
    - **When** the uploaded file size is zero bytes or format is unsupported (non-JPEG/PNG)
    - **Then** the system flags an error prompt "Please upload a valid image format" and keeps the milestone status as "Pending".

Story ID: US-05 — Adverse Weather Parametric Insurance Claim Trigger

- **User Story:**
  As a Smallholder Farmer facing severe climate damage,
  I want to submit a quick parametric weather claim through the app,

So that localized rainfall anomaly records trigger an accelerated relief payout without waiting for manual inspections.

- **Acceptance Criteria (Gherkin Syntax):**
  - **Scenario 1: Automated approval based on precipitation threshold**
    - **Given** local weather station telemetry exceeds the defined rainfall threshold (e.g., continuous localized heavy rain)
    - **When** the farmer files an incident report specifying affected crop acreage
    - **Then** the claims engine validates against regional climate indices, sets the claim status to "Pre-Approved", and queues micro-indemnity compensation for disbursement.
  - **Scenario 2: Discrepancy requiring manual assessor audit**
    - **Given** the farmer submits an adverse weather claim
    - **When** regional weather records show normal conditions within standard ranges
    - **Then** the system marks the claim as "Flagged for Assessor Review" and forwards the case file to the System Admin/Assessor queue.

Story ID: US-06 — Delivery Inspection & Escrow Fund Settlement

- **User Story:**
  As an Enterprise Buyer or Collection Center Representative,
  I want to confirm received weight and produce quality grading upon physical delivery,
  So that remaining escrow balances are securely released to the farmer.
- **Acceptance Criteria (Gherkin Syntax):**
  - **Scenario 1: Full acceptance and final payment settlement**
    - **Given** the farmer brings the harvested batch to the designated regional hub
    - **When** the buyer inspector inputs the confirmed weight and selects "Grade A / Accepted"
    - **Then** the contract updates to "Settled / Closed", and the escrow balance is automatically transferred to the farmer's registered account.
  - **Scenario 2: Grade dispute and partial escrow holding**
    - **Given** the produce delivered exhibits quality degradation below the contract specification
    - **When** the inspector rejects the batch or enters "Grade Dispute"
    - **Then** the system pauses automatic payout, marks the contract as "Under Dispute Review", and notifies the Platform Moderator.

3. Non-Functional Requirements (NFRs)

- **NFR-01 (Performance & Response Time):**
  - The mobile and web API endpoints must respond within **< 1.5 seconds** under standard 3G/4G rural network connections for all core data submissions.
- **NFR-02 (Data Security & Compliance):**
  - All transaction records, user authentication data, and escrow balances must be encrypted in transit via **TLS 1.3** and at rest using **AES-256** encryption.

- **NFR-03 (System Availability):**
    - The core platform services (API Gateway and PostgreSQL database) must maintain **99.5% uptime** during operational hours to support live auction and harvest logging.
- **NFR-04 (Usability & Accessibility):**
    - The mobile farmer interface must feature high-contrast UI elements, simplified icon-driven flows, and support local Sinhala and Tamil language localization for low-digital-literacy rural users.

4. Requirements Traceability Matrix (RTM)

Requirement ID	Business Objective / Scope Item	User Story Reference	Priority	Verification Method
REQ-01	Land & Cultivation Registry	US-01	High	User Acceptance Testing (UAT)
REQ-02	Forward Supply Contracts with Price Floor	US-02	Critical	Automated Unit & Integration Tests
REQ-03	Escrow Balance Commitment	US-03	Critical	Financial Gateway Simulation
REQ-04	Photo-Based Growth Milestone Logging	US-04	Medium	Mobile Functional Testing
REQ-05	Parametric Climate Claim Automation	US-05	High	Weather Station API Mock Testing
REQ-06	Quality Inspection & Final Settlement	US-06	Critical	End-to-End Workflow Validation